System Analysis and Design

Technical and Creative Question Bank with Model Answers — Weeks 9 & 10 (5 marks each)

Week 9: Capacity Estimation, Clustering and Load Balancing

Q1 (Technical, 5 marks)

List and explain any five key metrics used in capacity estimation.

Answer:

- **Throughput** (1): transactions or requests the system handles per second.
- **Latency** (1): time taken to complete a single transaction or request.
- **Response time** (1): total time a user waits for a response, including processing and queuing.
- **Load and concurrency** (1): number of users or operations active at the same time.
- **Utilization** (0.5): percentage of capacity currently used.
- **Business impact** (0.5): cost of over-provisioning against the risk of under-provisioning.

Full marks need five metrics, each with a correct explanation.

Q2 (Technical, 5 marks)

Describe the steps involved in capacity estimation. Explain the three parts of "Model the System".

Answer:

- **Define requirements** (0.5): identify expected workload, peak traffic and availability needs.
- **Analyze historical data** (0.5): find patterns and trends in past system data.
- **Model the system** (2):
  - *Workload modelling*: characterize the type and intensity of workloads, e.g. read-heavy vs write-heavy.
  - *Resource consumption modelling*: quantify CPU, memory and disk I/O used by each workload.
  - *Concurrency and scaling factors*: account for concurrent users and how each resource is affected.
- **Conduct load testing** (0.5): run stress and load tests to validate the model and find bottlenecks.
- **Provision resources** (0.75): calculate resources for projected capacity, with a margin for peaks.
- **Estimate growth** (0.75): forecast workload growth from business expectations.

Q3 (Creative / Scenario, 5 marks)

"ShopEasy", a startup, runs a flash sale. Last year its single server crashed within minutes. As the system analyst, explain how clustering and load balancing would prevent this. Link each benefit to the scenario.

Answer:

- **Clustering** (2.5): several servers (nodes) work together as one system.
  - *High availability*: if one node fails during the sale, others take over, so the shop stays online.
  - *Scalability*: extra nodes can be added before the sale to serve more shoppers without slowing down.
  - *Fault tolerance*: the cluster keeps running even when individual nodes fail, so one crash no longer takes the whole shop down.
  - *Resource management*: work is spread across nodes, so no single node becomes the bottleneck.
- **Load balancing** (2.5): incoming traffic is spread across the servers.

- *Efficient resource utilization*: no server is overwhelmed while others sit idle.
- *Improved performance*: pages load faster, so customers don't abandon their carts.
- *Redundancy*: if a server fails, traffic is redirected to healthy ones.
- *Scalability*: new servers can be added to the pool as traffic grows.

Conclusion: the single point of failure is removed and the sale can handle sudden traffic spikes.

Q4 (Creative / Calculation, 5 marks)

A hospital appointment portal normally gets 200 requests/sec, but 1,000 requests/sec at peak (e.g. results day). Each server can handle 250 requests/sec. Management wants a 20% safety margin. Using the capacity estimation steps, calculate the servers needed and justify your design.

Answer:

- **Define requirements and analyze data** (1): the peak of 1,000 req/s is the design target, not the average of 200 req/s.
- **Apply the margin** (1): 1,000 × 1.20 = **1,200 req/s** required.
- **Calculate servers** (1.5): 1,200 ÷ 250 = 4.8, rounded up to **5 servers**.
- **Utilization check** (0.5): at peak, 1,000 ÷ (5 × 250) = 80% utilization, which leaves headroom.
- **Justification** (1): add a load balancer to share traffic and cluster the servers so the portal survives a node failure. Confirm with load testing and revisit as growth is forecast. Under-provisioning risks a crash when patients need the portal; over-provisioning wastes money (business impact).

Optional extension: if the portal must keep running after one server fails, add a sixth server.

Week 10: System Development Life Cycle (SDLC)

Q1 (Technical, 5 marks)

Define SDLC and explain its five phases.

Answer:

- **Definition** (1): SDLC is a conceptual model of the policies and procedures for developing or altering systems throughout their life cycles.
- **Planning / feasibility study** (0.8): define the problem and scope, set objectives, confirm feasibility and produce a schedule. Output: feasibility report.
- **Analysis and specification** (0.8): gather, analyze and validate information; define and prioritize requirements. Output: Software Requirement Specification (SRS).
- **System design** (0.8): design the application, network, database, user interfaces and system interfaces, turning the SRS into a logical structure. Output: design document.
- **Implementation** (0.8): write the code, integrate modules, test, and install the system in its environment.
- **Maintenance / support** (0.8): provide user support, make changes, add new requirements and fix residual errors after deployment.

Q2 (Technical, 5 marks)

Name the document or deliverable produced at the end of each SDLC phase. Explain its purpose.

Answer:

Phase	Deliverable (marks)	Purpose
Planning	Feasibility report (1.25)	Confirms the whole project is viable; records threats, constraints, integration and security issues.
Analysis	SRS document (1.25)	Specifies the software, hardware, functional and network requirements.
Design	Design document (1.25)	Gives detailed specifications, used by later phases, that meet the SRS.
Implementation	Test report (1.25)	Lists errors found through the test plan (test cases, testing criteria, resource allocation).

Q3 (Creative / Scenario, 5 marks)

A university wants a new Library Management System. Show how you would apply the SDLC phases to this project. Give one specific activity for each phase.

Answer:

(1 mark per phase; any reasonable activity linked to the phase is accepted)

- **Planning:** define the problem (manual book records cause delays), set the objective (online search and borrowing), estimate cost, time and security risks. Produce a feasibility report.
- **Analysis:** interview students, teachers and librarians; prioritize requirements such as book search, due-date alerts and fines. Write the SRS.
- **Design:** design the database (books, members, loans), user interface and network setup. Create a maintenance and training plan and prepare the design document.
- **Implementation:** code the modules, integrate and test them, fix errors found, then install the system and train staff.

- **Maintenance:** provide help-desk support, fix bugs that appear after launch, and add new features (e.g. e-book lending) when requested.

Q4 (Creative / Analytical, 5 marks)

A food-delivery app was launched with no plan for support. Within weeks, users report bugs and new requirements keep arriving. Explain what the team failed to plan for in the SDLC, and what the maintenance phase should include.

Answer:

- **Diagnosis** (1): the team treated the SDLC as ending at launch. The Maintenance/Support phase, and the design phase's "contingency, training, maintenance and operation plan", were ignored.
- **Maintenance activities** (3):
  - user support, by phone or on-site, once the system is installed
  - implementing changes the software needs over time, and new requirements after deployment
  - handling residual errors that remain even after the testing phase
- **Scale** (0.5): large systems need support for longer than small ones.
- **Recommendation** (0.5): plan support during the Design phase and reserve time and budget for it in Planning.

Source: System Analysis and Design lecture slides, Week 9 and Week 10 (Dept. of CSE, University of Global Village).